# A Modular Pipeline for WNBA Team and Player Performance Forecasting: Architecture, Validation, and Threats to Validity

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We describe the architecture, statistical methodology, and validation results of a modular forecasting pipeline for the Women's National Basketball Association (WNBA), adapted from an earlier NBA-focused system. The pipeline ingests possession- and shot-level play-by-play data, fuses adjusted plus-minus (RAPM), box-score, and Elo-style signals into team and player ratings, and propagates these ratings through market-calibration, player-prop-pricing, playoff-simulation, and bankroll-risk layers. We report leave-one-season-out cross-validated performance for the primary win-probability model (pooled accuracy 0.639, 95% CI 0.618-0.661) and for a market-line-fitting sub-model (pooled out-of-fold MAE 13.11 points, 95% CI 12.70-13.52), and benchmark the former against a published WNBA-specific model. We give particular attention to a class of defects discovered during a structured remediation review – a home-court-advantage parameter transcription error, silent no-op pipeline stages that produced misleadingly "complete" logs, and an identifier-mismatch join collision between two internal player-rating tables – because each is a general failure mode for any adapted or rapidly-iterated sports-analytics codebase, and because one of them (the transcription error) appears to have been self-validated by a downstream optimizer fit on its own contaminated residuals. We conclude with a discussion of unresolved architectural gaps and features carried over from the NBA source system without WNBA-specific recalibration.

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# 1 Introduction

Sports-forecasting systems that are ported from one league to a structurally different one – shorter games, shorter seasons, smaller rosters, a smaller and more volatile franchise universe – face a specific reproducibility risk: constants, thresholds, and architectural assumptions tuned for the source league can be carried into the target league silently, producing a system that runs without error but encodes the wrong league. This paper documents such a port: a modular WNBA forecasting pipeline derived from an existing NBA system, and a structured multi-phase review process undertaken specifically to surface exactly this class of defect before treating the WNBA system's output as trustworthy.

We organize the paper around the pipeline's actual execution order rather than around a idealized statistical taxonomy, because several of the most consequential findings in this project were *architectural* rather than *statistical* – for example, a pipeline stage that logged as "successfully completed" while performing no work at all (§5.2). A purely statistical description of the modeling choices would miss this entirely.

The system was developed against the historical design references `REPO_MAP.md` and `WNBA_REPLICATION_PLAN.md`, which describe the original NBA system and the initial WNBA port plan respectively. We treat those two documents as historical background only; where the current implementation materially diverges from them, we say so explicitly (§7), since several components planned as long-term synthetic placeholders (RAPM, shot quality) were subsequently built out against real data, and the port surfaced at least one unanticipated volatility risk (WNBA franchise count changing from 12 to 15 teams across the seasons studied) exactly as the planning document had flagged as a possibility.

# 2 Data Sources

The pipeline's primary data source is the `wehoop` R package, an ESPN- and WNBA-Stats-API-backed data provider analogous to the NBA-focused `hoopR` package. Three feed types are consumed:

- **Team and player box scores** (`wehoop::load_wnba_*_box()`), 2019-2026, used for Elo updates, box-score-derived plus-minus, rotation/minutes modeling, and reconciliation of logged predictions against real outcomes.
- **Possession-level play-by-play** (`wehoop::load_wnba_stats_possessions()`), 292,569 rows across the same period (253,970 retained after removing seven rows with implausible point totals), used to build the possession-level design matrix underlying the RAPM model (§4.1).
- **Shot-location data** (`wehoop::load_wnba_stats_shots()`), 20,000-42,000 shots per season across all eight seasons studied, used for expected-points and shooting-luck estimation (§4.4).

A fourth, auxiliary source is ESPN's roster and injury endpoints, scraped directly rather than through `wehoop`, and a fifth is a live sportsbook-odds feed (The Odds API) used for market calibration (§4.6).

## 2.1 Data-quality corrections

Several corrections to the raw feeds were necessary before any of the above data could be used without silently corrupting downstream aggregates. We highlight three:

**Franchise universe instability.** The WNBA expanded from 12 franchises (2019-2024) to 13 (2025, Golden State Valkyries) to 15 (2026, Toronto Tempo and Portland Fire). Every structural constant that depends on the team count – schedule length assumptions, standings simulation, the RAPM team-ID map – reads from a single centralized registry rather than a hardcoded literal, specifically to absorb this churn without requiring code changes at every site that iterates over teams.

**Non-franchise rows and team-code normalization.** Raw box-score data includes All-Star and exhibition rows under season-varying non-franchise codes (`DEL`, `WIL`, `ALL`, `USA`, `STE`, `CLA`, `COL`, `COOP`, `SPO`, and others). These must be filtered by restricting to the canonical franchise code list – but the underlying data source uses the code `CONN` for the Connecticut Sun in seasons 2019-2023, while the canonical registry uses `CON`. Applying the exhibition-row filter *before* normalizing `CONN` to `CON` silently discards five entire seasons of a genuine franchise's data rather than only the intended exhibition rows – a defect that produces no error, no missing-file warning, and a plausible-looking (merely smaller) dataset, and is therefore not the kind of bug that reveals itself through ordinary testing. It was caught only by an explicit season-by-season row-count and non-franchise-code audit.

**Injuries endpoint schema divergence.** The WNBA injuries endpoint, unlike the structurally similar NBA endpoint that most of the injury-scraping code was originally written against, has no `athlete.id` field. The player identifier must instead be parsed out of the href of an embedded "Player Card" link – a source-specific workaround with no general analogue, included here because it is a concrete example of the broader risk of assuming schema parity between two leagues' otherwise similar API surfaces.

## 3 Pipeline Architecture

The pipeline is orchestrated by a single driver script that sources each stage script in dependency order inside a fault-isolating wrapper: a stage failure is logged and the run proceeds rather than aborting the entire pipeline, consistent with a fail-safe design principle applied throughout the codebase (every consumer of an external or possibly-stale resource falls back to a clearly labeled synthetic value rather than halting). As of the most recent verified full run, all eighteen stages completed with zero failures in under forty seconds of wall-clock time.

The stage sequence, in execution order, is: a pre-flight cache/schema health check; incremental box-score ingestion; roster and injury synchronization (parallel with a team-style-embedding subprocess); reconciliation of the previous day's logged predictions against real outcomes; three independent feature-engineering stages executed in parallel (shot quality, fatigue/travel, RAPM); team-rating fusion; market-odds generation and same-day prediction logging; two independent player-prop pricing stages executed in parallel (a parametric count model and a copula-based multivariate model); playoff-series Monte Carlo simulation; bankroll risk-state computation; and finally notification dispatch and ledger rollup.

Two design choices in this orchestration merit comment as methodological, not merely implementation, decisions. First, each stage's declared output cache file is checked for an advanced modification time as a post-condition, not merely for its existence – this distinction is what later exposed several silently non-functional stages (§5.2). Second, downstream stages consume upstream stages' outputs exclusively through versioned cache files on disk rather than through in-process object passing, which makes every intermediate artifact of the pipeline independently

inspectable and re-runnable, at the cost of requiring the freshness-checking discipline just described.

# 4 Statistical and Modeling Methodology

## 4.1 Adjusted plus-minus (RAPM)

Player value at the possession level is estimated via regularized adjusted plus-minus. A sparse possession-by-player design matrix is constructed with separate offensive and defensive indicator columns per player, and a weighted ridge regression is solved with a fixed penalty ($\lambda = 900$) selected from a small grid search ($\{50, 100, 300, 1000\}$) rather than by cross-validation. We flag this explicitly as a limitation: the ridge penalty governing how much possession-level noise is shrunk toward league-average is, at present, chosen by inspection rather than by an out-of-sample criterion, and a future revision should replace this with a proper cross-validated selection. Numeric team identifiers from the possession-level API are mapped by hand to the pipeline's canonical franchise codes. Where a possession-level pull fails or returns insufficient data, a box-score-derived fallback – a minutes-weighted, shrunk plus-minus estimate – is substituted, and every downstream consumer of RAPM output can distinguish real from fallback estimates via an explicit data-source flag, along with per-player possession counts that allow low-sample players to be down-weighted or excluded.

## 4.2 Team rating fusion

Team strength is estimated as a weighted fusion of four independent signals: an Elo rating updated game-by-game from box scores with a fixed K-factor and a fractional offseason regression toward the mean; a possession-weighted team aggregate of the RAPM ratings above; a box-score-derived player-level box plus-minus metric, itself a fifty-fifty blend of a box-score-creation formula and a shrunk empirical plus-minus, restricted to players with a minimum minutes threshold and current-season performance up-weighted relative to prior seasons; and a top-rotation-weighted live team rating that combines historical minutes distribution with player ratings and applies injury-driven minute reallocation. The four signals are combined into a single matchup-level fair margin with a fixed linear weighting, add a home-court-advantage term, and are converted to a win probability via a normal approximation whose standard deviation is set to the empirical standard deviation of real home-margin outcomes (1,953 games) rather than to a value carried over from the source league.

## 4.3 Shot quality

Team-level shooting efficiency is decomposed into an expected-points-per-shot baseline, computed from real shot-location data via a distance-bucketed field-goal-percentage curve, and

an observed-minus-expected "shooting luck" residual. The expected-shooting curve itself was carried over from the source league's system without WNBA-specific re-fitting. We consider this a live limitation rather than a resolved design choice: thirteen of fifteen teams in the current output show a *negative* shooting-luck residual, which is difficult to interpret as thirteen teams simultaneously underperforming a correctly calibrated baseline, and easier to interpret as evidence that the baseline curve itself runs systematically "hot" for the league's actual shot-distance and efficiency profile. A per-team synthetic fallback is used only for teams with fewer than three real games in a given window.

### 4.4 Fatigue and travel

An arena database of geographic coordinates, time zone, and altitude for every current franchise venue feeds a haversine-distance travel-fatigue model with additive penalties for travel distance, circadian time-zone shift, altitude, and back-to-back scheduling. The underlying arena data is real and current (including the two 2026 expansion franchises); the altitude penalty is included for structural completeness but in practice never triggers, as no current WNBA market clears the model's altitude threshold. We note as a limitation that the schedule against which this component is demonstrated in the pipeline's own test harness is a small fabricated fixture rather than a live schedule feed, distinguishing "the underlying model uses real geography" from "the demonstrated output reflects a real slate."

### 4.5 Market calibration

Fair game margins are computed as the sum of the team-rating differential (from §4.2), the home-court-advantage constant, a fatigue-driven adjustment, and a star-power delta computed from each team's highest-RAPM player. Win probabilities are obtained via the same normal approximation described in §4.2. When a live odds feed is unavailable, a clearly labeled synthetic six-game fixture is substituted and every downstream consumer can distinguish real from synthetic market data via an explicit source tag. Leave-one-season-out cross-validation of the rating-input dampening and clamping parameters governing this stage yields a pooled out-of-fold mean absolute error of 13.11 points (95% CI 12.70-13.52) under the current configuration, an improvement over a pooled MAE of 13.57 under the system's pre-optimization default configuration.

### 4.6 Player and combination props

Individual player statistical props are priced using a zero-inflated negative binomial distribution with per-statistical-category dispersion and zero-inflation parameters re-estimated empirically on WNBA data. Opponent adjustment is applied via an empirically fitted slope relating a player's expected mean to the opponent's defensive rating, clamped to a bounded multiplicative range to prevent extrapolation beyond the data used to fit it. Multi-statistic combination props

(for example, points plus rebounds plus assists) are priced via Monte Carlo simulation from a multivariate Gaussian copula whose pairwise correlation structure between points, rebounds, and assists is estimated directly from real WNBA rotation data rather than carried over from the source league.

### 4.7 Playoff simulation

Playoff series outcomes are simulated via a format-aware Monte Carlo routine that encodes the current WNBA playoff format for each round, including a best-of-three first round in which, unlike a naive higher-seed-hosts-more assumption, the *lower* seed hosts the second game. Team strength inputs for simulation apply an additional weighting to a team's top RAPM contributors to reflect increased reliance on top players in a shortened playoff rotation.

### 4.8 Risk management and live calibration

A quarter-Kelly staking framework with drawdown-triggered de-risking (staged reductions at three- and five-unit drawdowns, and a full trading halt at eight units) governs recommended bet sizing, computed against a real resolved-bet ledger where available. A separate live-calibration module implements standard probabilistic scoring (Brier score, Brier skill score, log-loss) correctly parameterized for the WNBA's forty-minute game length rather than the source league's forty-eight-minute length; however, we report this module as validated only against a small synthetic demonstration feed, since no live in-play odds integration has yet been wired to a real endpoint, and no real-world calibration numbers for in-game win probability currently exist for this system.

## 5 Validation Results

The primary win-probability model was evaluated via leave-one-season-out cross-validation, yielding a pooled accuracy of 0.639 (95% CI 0.618-0.661). Benchmarked against a published WNBA-specific model (Alves and Barbosa 2025), which reports 67.48% accuracy, this represents a 3.58-percentage-point shortfall against an external reference point, which we report plainly as a gap rather than reframe favorably.

A separate hyperparameter sweep over the rolling-window feature size found a single-split peak (accuracy 0.645, AUC 0.678, Brier score 0.225) at a twelve-game window; we flag this result as descriptive rather than confirmatory, since it reflects one train/test split and has not been cross-validated.

Prior to a logging-schema correction described in §5.3, a dashboard-facing accuracy figure was computed over a prediction log in which the large majority of rows were synthetic diagnostic fixtures rather than real resolved games, yielding a misleadingly high headline accuracy figure.

Following the correction, the same query correctly returns zero real resolved predictions, since the log's real history begins only after the correction was deployed. We report this as an honest, if currently uninformative, result, and note that a home-court-advantage optimizer intended to re-derive the model's home-court constant from real resolved outcomes has, as of this writing, likewise never been run against a non-empty set of real resolved games.

# 6 Discovered Defects with Methodological Significance

A structured, multi-phase code review of this pipeline surfaced several defects whose significance is methodological rather than purely operational: each would have silently biased the pipeline's statistical output while producing no error and no visibly abnormal log. We describe three in detail because each represents a general failure mode relevant beyond this specific codebase.

## 6.1 A parameter transcription error in the home-court-advantage constant

The market-calibration stage's live code set its home-court-advantage input to 11.6 at the point of use, three lines below a code comment stating the intended value of 2.40 – apparently a value from the source NBA system's unrelated margin-standard-deviation parameter, mispasted into the wrong slot. The quantitative consequence is severe rather than cosmetic: holding two otherwise identical teams constant, a home-court-advantage input of 11.6 against the model's margin standard deviation of 14.0 implies a 79.6% home win probability, against the WNBA's actual observed home win rate of approximately 55-58%; the corrected value of 2.40 implies 56.8%, consistent with observation. Live evidence consistent with the bug being active in production was found directly: on the day it was discovered, all recommended bets in the day's slate were on the home side, all were pinned at the staking model's maximum bankroll-percentage cap, and several carried implausible expected-value figures in excess of 100%. The fix centralizes the constant in the single shared configuration module read by every consumer, and every cached artifact produced while the bug was live was identified and regenerated rather than left in place.

We highlight a second-order consequence of this defect in §7: a separate hyperparameter-optimization run, intended to independently re-derive the correct home-court-advantage value from residuals of already-logged predictions, recovered a value close to the buggy 11.6 rather than the correct 2.40 – because the residuals it was fit against were themselves generated by the buggy live system. An optimizer trained on the output of a broken instrument can reproduce and appear to validate the instrument's own error; we consider this a noteworthy general caution for any pipeline that retrospectively tunes parameters against its own historical predictions rather than against independently verified ground truth.

### 6.2 Silent no-op pipeline stages

Several pipeline stages gated their primary logic on a condition intended to detect "this script was invoked directly from the command line" – true only when the script is the top-level entry point of an `Rscript` process, and never true when the same script is instead `source()`d from within another script's function body, as every stage in this pipeline is when invoked by the orchestrator. The practical effect was that four stages, including the roster synchronization, injury scraping, team-rating fusion, and playoff-simulation stages, logged as completing successfully in well under a second while performing no actual work, leaving their output caches silently stale with modification timestamps dating from days before the "successful" run that claimed to have regenerated them. This is a defect class we consider particularly serious for any data pipeline with a similar orchestration pattern, because the log output – ordinarily the first and often only artifact inspected to confirm a pipeline ran correctly – was not merely incomplete but affirmatively misleading. The remediation replaces the invocation-detection condition with one that correctly distinguishes sourced execution from direct invocation, and additionally instruments the orchestrator itself to assert that each stage's declared output file's modification time actually advanced during that stage's execution, so that a future recurrence of this failure mode would be caught structurally rather than relying on a human noticing an anomalously fast stage.

### 6.3 An identifier and naming mismatch between two rating tables

The pipeline's possession-level RAPM output and its player-roster cache use two incompatible player identifier schemes – RAPM's numeric identifier is drawn from a different numbering system than the roster cache's identifiers, and RAPM's player-name field stores only a surname where the roster cache stores a full name. A join on either field independently returns zero matches, which is at least immediately visible; the more dangerous case is a composite join key of team and normalized surname, which resolves the great majority of players correctly but is not itself collision-free, since a number of team-surname combinations correspond to more than one real player. A shared, single join utility now resolves each such collision deterministically by total possession volume with an explicit tiebreak rule, and is the exclusively sanctioned path for this join anywhere in the codebase, rather than leaving each consuming script to reimplement the join – and its collision handling – independently.

## 7 Discussion: Divergence from the Original Design Plan and Remaining Gaps

The system's original port plan anticipated that several major modeling components – adjusted plus-minus, shot quality, fatigue and travel modeling, playoff simulation, bankroll risk management, and combination-prop pricing – would remain long-term synthetic placeholders,

ported chiefly for interface compatibility with the source system rather than as fully realized WNBA-specific models. This assumption did not hold for the two most statistically central of these components: both the adjusted-plus-minus and shot-quality engines were subsequently built out into models operating on real possession- and shot-location-level WNBA data, a substantial expansion of scope beyond what was originally planned. Fatigue and travel modeling occupies an intermediate state – real venue geography feeding a still-fabricated demonstration schedule – and should be read accordingly.

Separately, several thresholds and structural assumptions calibrated for the source league's longer games and larger rosters were carried over into the WNBA system unchanged and have not yet been independently validated against WNBA data: a fixed eight-player rotation size, a thirty-minute co-occurrence threshold used to detect star-player lineup overlap, and, as discussed in §4.4, the shot-quality baseline curve itself. We regard each of these as a candidate for WNBA-specific recalibration in future work, distinct in kind from the outright defects described in §5, since each runs without error and produces plausible-looking output – the concern is calibration, not correctness of implementation.

Finally, we note one architectural inconsistency that remains unresolved at the time of this writing and was not part of the enumerated defect list addressed during the remediation phases described above: the team-style-embedding subprocess currently writes its output into a database file, while the downstream visualization component that consumes team embeddings still expects a flat CSV file at a different path. This should be treated as a live gap rather than a historical finding, and any reported similarity between team playing styles derived from the embeddings dashboard tab should be read with the additional caveat that the embedding vectors themselves were originally trained on fabricated, rather than real, matchup data.

## 8 Conclusion

We have described a modular WNBA forecasting pipeline, its constituent statistical models, and the results of a structured review process that surfaced several defects of genuine methodological consequence rather than mere code-quality issues. The pipeline's primary win-probability model achieves a cross-validated accuracy of 0.639, trailing a comparable published benchmark by 3.58 percentage points, and several of its components – most notably shot quality and live in-play calibration – carry explicit, unresolved calibration or implementation gaps that should temper confidence in their output until addressed. We consider the discovered defects, and in particular the home-court-advantage transcription error and its apparent self-validation by a downstream optimizer, to be of interest beyond this specific system, as instances of general failure modes – silent parameter corruption, misleading success logging, and identifier-mismatch join collisions – that can affect any actively developed, multi-stage sports-analytics or forecasting pipeline.

# 9 References

Alves, D., and R. Barbosa. 2025. "Predicting WNBA Game Outcomes Using Machine Learning Models." *Computation* 13 (10): 230. https://doi.org/10.3390/computation13100230.